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Conceptual analysis of cathode exhaust gas recirculation to reduce idling power and enable faster freeze starts in polymer electrolyte membrane fuel cell systems

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ABSTRACT

Proton exchange membrane (PEM) fuel cells are increasingly recognized as a viable clean energy option in mobility, due to their high efficiency and minimal greenhouse gas emissions. A significant challenge in enhancing PEM fuel cell systems lies in improving the efficiency of the air processing subsystem (APS). Despite recent progress, further optimization is imperative, particularly in addressing costs, service life, efficiency, and dynamic behaviour. This study presents an innovative approach to cathode exhaust gas recirculation (CEGR) in the fuel cell's air supply system, enhancing both lifetime and efficiency by optimizing idling and freeze-start behaviours, thus enhancing the operational range. The system recirculates oxygen-depleted humid exhaust gas to the fuel cell stack, facilitated by components including a control valve, water separator, collection tank, and diaphragm pump. Simulation models provide detailed correlations between the gas composition of recirculated exhaust gas, oxygen stoichiometry, and cell voltage. Based on these insights, a control system for freeze starts and idling is developed. Simulation findings highlight the potential of the exhaust gas recirculation system to save up to 83 % hydrogen and extend service life during idle power through targeted cell voltage reduction. Furthermore, additional heat generation enables freeze starts up to 4.6 times faster at temperatures below −30 °C.

ABBREVIATIONS			
AP	Air processing	δ	MEA component thickness
BoP	Balance of plants	E^0	Standard potential
BPP	Bipolar plate	E_N	Nernst potential
cat	Cathode	F	Faraday constant
CEGR	Cathode exhaust gas recirculation	i_{int}	Internal current density
CFD	Computational fluid dynamics	i_{lim}	Limiting current density
Cons	Consumed	i_0	Exchange current density
DC	Direct current	I	Stack current
DCDC	DC-to-DC converter	λ_{air}	Air stoichiometry
Eff	Effective	λ_{O2}	Oxygen stoichiometry
FC	Fuel cell	$\dot{m}$	Mass flow
GDL	Gas diffusion layer	M	Molar mass
Max	Maximal	N	Stack cells
MEA	Membrane electrode assembly	OCV	Open circuit voltage
min	Minimal	P	Power
PEM	Proton exchange membrane	p^0	Standard pressure
		p_i	Partial pressure of the reactant i
PEMFC	Proton exchange membrane fuel cell	R	General gas constant
Req	Required	s	Iteration step
red	Reduction reaction	σ	Specific conductivity
Rec	Recirculated	t	Time
PID	Proportional-integral-derivative	T	Temperature
Sup	Supplied	U	Cell voltage
		X_{EGR}	Recirculation ratio
SYMBOLS		ξ_1	Pre-logarithmic correction factor
A	Nominal active area	y	Volume fraction
$\alpha_{i\ (ox./red)}$	Charge transfer coefficient	z	Valency

(*continued on next column*)

1. Introduction

The low-temperature proton exchange membrane fuel cell (PEMFC) stands out as a leading technology in the quest to decarbonize the

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mobility sector [1–3]. Renowned for its environmental benefits, impressive power density, and exceptional efficiency, the PEMFC has already gained widespread adoption. Nevertheless, the restricted freeze start behaviour [4–7] and limited useful lifetime [8–10] pose challenges for PEMFC-powered vehicles, constraining their potential applications.

The limited freeze start performance challenges the use of fuel cell vehicles in winter and in freeze zones. Optimizing the freeze start, which involves starting the FC at temperatures below freezing and reaching the operating temperature, would expand the range of applications for the PEM fuel cell [11,12]. The Department of Energy (DOE) aims therefore for a self-start capability at −30 °C, an assisted start at −40 °C, and achieving 50 % of the rated output power within 30 s at −20 °C [13].

Moreover, the durability of the PEM fuel cell is a prerequisite for a commercial breakthrough. Fuel cells utilized in vehicles endure severe conditions, including sustaining open circuit voltage (OCV), idling, handling dynamic loads, start-up and shutdown procedures, operating at maximum power, managing overload situations, and enduring freeze-thaw cycles [14–16]. To achieve the target useful lifetime of 8000 h for light-duty vehicles and 30,000 h for heavy-duty trucks, it is therefore necessary to avoid these harsh operating conditions and mitigate their effects [17].

This study explores the innovative concept of cathode exhaust gas recirculation (CEGR) to enhance freeze start performances of PEM fuel cells and minimize their idle power. To our knowledge, there is no known study that designs a complete CEGR system for mobility applications, including a possible operating strategy, control model, and risk analysis, while demonstrating the optimization potential for freeze starts and idling through simulation. This study attempts to close this research gap.

In general, a CEGR system integrated into the air supply system of the fuel cell extracts warm, oxygen-depleted, and humid air from the exhaust gas path and feeds it into the fresh air mass flow. This increases the temperature and relative humidity and reduces the oxygen concentration of the stack supply air.

Kim et al. [18] analyse the CEGR for self-humidification and describe two basic options for system architecture with CEGR. Self-humidification can be used to eliminate the need for an external humidifier, thereby reducing installation space, costs, mass, and system complexity. The system architecture options include recirculation of the exhaust gas before the air compressor with a control valve and recirculation of the exhaust gas after the air compressor with an additional recirculation compressor.

The option with a control valve has the advantage of a simpler design [18,19]. Recirculation with a recirculation compressor is associated with less condensation when the recirculation air meets the fresh air and therefore offers advantages in self-humidification [20,21].

Rodosik et al. [21] confirm experimentally that sufficient self-humidification can be achieved by CEGR. Although this is associated with a permanent reduction in cell voltage due to the lower oxygen concentration.

Zhang et al. [20] focus, in addition to self-humidification, on high potential limitation and are able to keep the single cell voltage below 0.8 V at low currents.

In the literature, a cell voltage of 0.8 V is specified as the threshold value for voltage-dependent degradation and limits therefore the minimal idling power. Above this voltage limit, there is increased degradation due to chemical ionomer degradation, chemical carbon corrosion and platinum degradation. This voltage limit can occur both when idling and during start-stops [14,22].

Zhang et al. [20] also describe the water removal function with CEGR, which prevents flooding of the stack and allows oxygen to be better distributed along the flow channels.

Liu et al. [23] conduct comprehensive measurements using electrochemical impedance spectroscopy (EIS) and distribution of relaxation times (DRT) with CEGR at low currents across various recirculation levels. This analysis effectively elucidates the polarization curve, considering factors such as oxygen diffusion, oxygen reduction reaction, and proton transport effects. Their findings confirm that CEGR positively impacts the durability at low current.

With regard to freeze start optimization, Toyota [24] presents a concept for rapid start-up, which increases the concentration overvoltage by limiting the supply air, reducing thereby the cell efficiency and thus generating more waste heat. However, reducing the cell voltage by limiting the supply air alone can lead to instability and increased degradation due to the lack of water removal and poorer oxygen distribution [25,26].

The importance of CEGR during a freeze start in this study is based on the potential that waste heat can be increased without additional degradation or instabilities and that additional heat can be recovered by recirculating the warm exhaust gas.

As described in our previous publication [27] and in Ref. [28], the presence of nitrogen in the cathode supply air of the stack lowers the oxygen partial pressure, which leads to a significantly increased concentration overvoltage and a drop in the stack voltage.

Nevertheless, it is still largely unexplored to what extent oxygen stoichiometry can be reduced to achieve a cell voltage reduction without irreversible degradation.

In general, an oxygen stoichiometry lower than one, a phenomenon termed cathode starvation or oxygen starvation, generates hot spots on the polymeric membrane, triggers polarity reversal, accelerates the degradation of the catalyst metal and causes corrosion of the carbon carrier [29–31]. However, oxygen starvation is not considered as critical for the cell degradation as hydrogen starvation [32].

Gerard et al. [33] confirm that oxygen starvation should be avoided at all costs during dynamic load jumps and high current densities, but emphasise that low oxygen stoichiometry is permissible under low current density and steady-state conditions.

Zamel et al. [34] show the characteristic comma-shaped polarization curves for oxygen stoichiometries below 1. Under stationary conditions, when the theoretically limited current density is reached and the cell voltage drops further, they observe a subsequent recovery of the current density in the oxygen-depleted regions.

Bodner et al. [32] conduct experimental research on the induced degradation due to air starvation in a PEMFC using accelerated stress tests. They observe an enhancement in power density and cathode-active catalyst area after 100 cycles with a 10 s air starvation period. However, they confirm that degradation emerged as the dominant phenomenon with longer starvation periods. A notably accelerated voltage decay rate is noted after 60 s.

This study proceeds under the assumption that reducing oxygen stoichiometry, without compromising the fuel cell's lifetime, is permissible under stationary conditions, for a limited duration and with the usage of CEGR, ensuring well-distributed oxygen. It is assumed that maintaining an oxygen stoichiometry above 1 during idling is advisable, given the prolonged operation of CEGR in this state. Conversely, an oxygen stoichiometry below 1 is considered acceptable during freeze starts, if oxygen starvation is kept well below 60 s.

The study initiates with an outline of the development methodology, followed by a theoretical overview of the calculation of gas composition and cell voltage. Subsequently, the results section delineates the system architecture, the relationship between gas composition and cell voltage, the potential during freeze starts and idling, and the control strategy. Towards the end of the results section, a risk analysis is conducted, leading to the conclusion of the paper.

2. Methodology

Fig. 1 delineates the methodological framework employed throughout the design process of the CEGR. This process unfolds across three primary phases: System Requirements, Conceptualization, and Development and Realization. The tools used are shown in the lower section of the illustration.

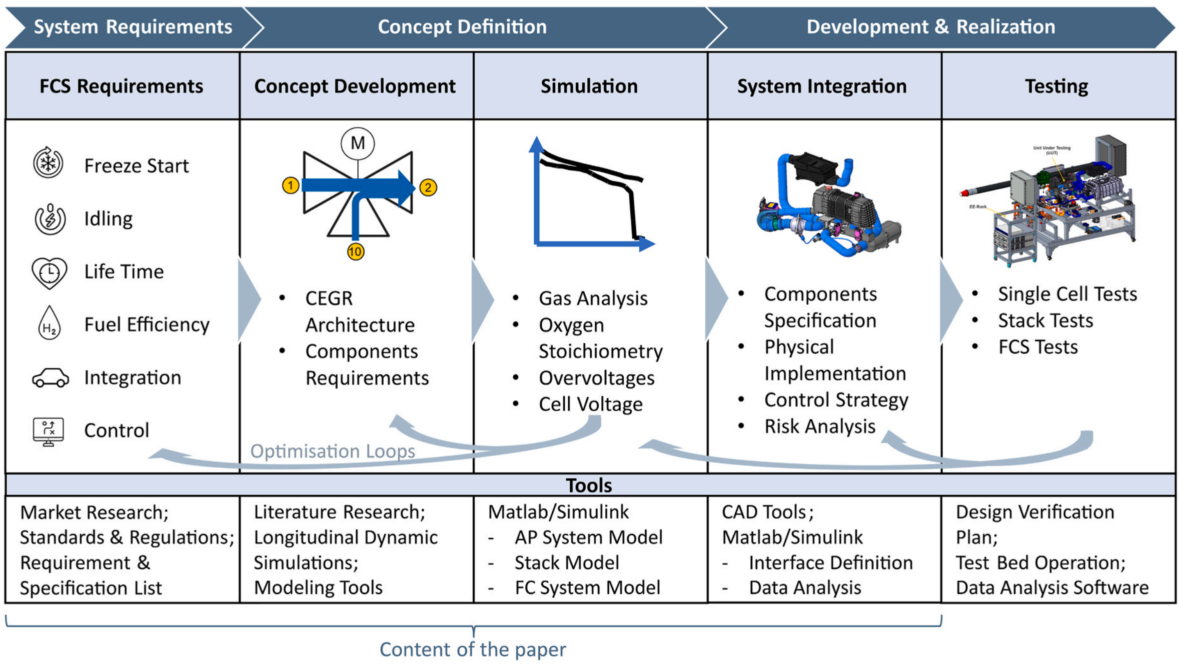

Fig. 1. Methodological framework employed throughout the design process.

Based on the system requirements, the most important specifications for the vehicle are identified. Emphasis is placed on critical parameters such as freeze start and idling within the designated operational domain. Furthermore, adherence to established standards about lifetime, fuel efficiency, system integration and controllability is deemed imperative.

Transitioning to conceptualization, the system architecture for the CEGR, encompassing all constituent elements, is formulated. This is followed by a simulation in which the gas composition and stack voltages are analysed. With the knowledge gained from these analyses, the system components are specified and integrated.

The final phase of the design process involves testing the entire system, including single cell-, stack- and FCS-testing. However, it is important to mention that testing procedures are not addressed within this study.

3. Theory and calculations

3.1. Gas composition

In this section, the methodology for calculating the oxygen demand, along with the gas composition, is elaborated. The following assumptions are considered.

1) 0D-quasi stationary calculations
2) Air, hydrogen, and vapour are considered as ideal gas with constant thermo-physical properties
3) Condensation takes place once the dew point is reached. No evaporation of liquid water takes place under the dew point.
4) The latent heat of condensation is included within the thermal balance

Calculation constants, along with system and component information, are presented in Table 1 for reference and analysis.

The air fed to the cathode $\dot{m}_{feed}$ is determined according to equation (1) by calculating the oxygen consumption $\dot{m}_{cons,O2}$ and multiplying it by the air stoichiometry λ_{air} and dividing it by the molar mass and volumetric fraction of air in oxygen. $\dot{m}_{cons,O2}$ is calculated according to Faraday's laws, which are given in equation (2).

Table 1
System information.

Variable	Symbol	Value	Unit
Stack current	I	5 to 560	A
Stack cells	N	359	/
Nominal active area	A	250	cm^2
Air stoichiometry	λ_{air}	4.5 to 1.8	/
Oxygen molar mass	M_{O2}	31.99880	g/mol
Nitrogen molar mass	M_{N2}	28.01340	g/mol
Dry air molar mass	M_{Air}	28.96440	g/mol
Faraday constant	F	96485.33212	C/mol
Volume fraction of oxygen in dry air	y_{O2}	0.20946	/
Volume fraction of nitrogen in dry air	y_{N2}	0.78084	/

$$\dot{m}_{feed} = \dot{m}_{cons,O2} \cdot \lambda_{air} \cdot \frac{M_{Air}}{y_{O2} \cdot M_{O2}} \tag{1}$$

$$\dot{m}_{cons,O2} = \frac{I \cdot N}{4 \cdot F} \cdot M_{O2} \tag{2}$$

The oxygen stoichiometry λ_{O2} is introduced in equation (3), which represents the ratio of oxygen fed $\dot{m}_{feed,O2}$ to oxygen consumed $\dot{m}_{cons,O2}$.

$$\lambda_{O2} = \frac{\dot{m}_{feed,O2}}{\dot{m}_{cons,O2}} \tag{3}$$

$\dot{m}_{feed}$ remains constant throughout the subsequent calculations for the respective current, see equation (4) and is irrespective of CEGR.

$$\dot{m}_{feed} = f(I) \tag{4}$$

To integrate CEGR into the calculations and facilitate better scaling, a characteristic parameter called the recirculation ratio X_{EGR} [%] is defined in equation (5). This ratio represents the proportion of dry recirculated mass flow $\dot{m}_{rec}$ to the total dry cathode mass flow $\dot{m}_{feed}$, see Fig. 2.

$$X_{EGR} = \frac{\dot{m}_{rec}}{\dot{m}_{feed}} \cdot 100 \tag{5}$$

The recirculation ratio X_{EGR} provides insight into the amount of low-oxygen exhaust gas that is being recirculated. Variations in stack

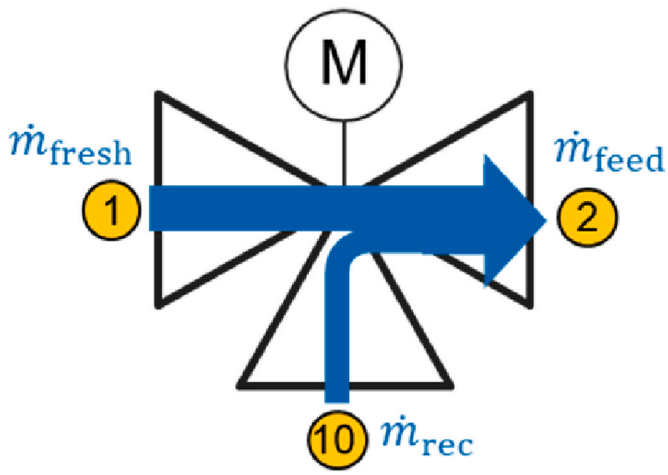

Fig. 2. Recirculation valve and the definition of the mass flows.

operating conditions resulting from CEGR can be effectively represented based on this ratio.

With CEGR, a mixture of oxygen-depleted air is fed into the fresh air path of the PEMFC, resulting in a decrease in oxygen and an increase in nitrogen concentration. This necessitates an iterative calculation process, as the recirculated air is interdependent with the air entering the stack, and vice versa. The iterative calculation begins with the specified total mass flow $\dot{m}_{\text{feed}}$ and the defined recirculation ratio, X_{EGR}. These parameters enable the determination of gas composition at a particular recirculation ratio based on the starting conditions with ambient air. With each iteration, the oxygen content diminishes while the nitrogen concentration rises. This process repeats until the difference in gas concentrations compared to the previous iteration step is less than 1/1000. Typically, stable gas concentrations are achieved after approximately 13 iteration steps with a fixed-pointed method, indicating convergence of results. The iterative calculation equation for the mass flow of the supplied oxygen $\dot{m}_{\text{feed,O2}}$ and the supplied nitrogen $\dot{m}_{\text{feed,N2}}$ can be described using equations (6) and (7), where s denotes the iteration step. In these simplified calculations, the vapour content contained in the air and other gases are neglected. The primary focus is on the concentration ratio between oxygen and nitrogen, as these are the main components that significantly impact the performance and efficiency of the system.

$$\dot{m}_{\text{feed,O2}}(s) = \frac{X_{\text{EGR}}}{100}\cdot\left[\dot{m}_{\text{feed,O2}}(s-1) - \dot{m}_{\text{cons,O2}}\right] + \dot{m}_{\text{feed}}\cdot\left[1-\frac{X_{\text{EGR}}}{100}\right]\cdot\frac{M_{\text{O2}}\cdot y_{\text{O2}}}{M_{\text{air}}} \tag{6}$$

$$\dot{m}_{\text{feed,N2}}(s) = \frac{X_{\text{EGR}}}{100}\cdot\dot{m}_{\text{feed,N2}}(s-1) + \dot{m}_{\text{feed}}\cdot\left[1-\frac{X_{\text{EGR}}}{100}\right]\cdot\frac{M_{\text{N2}}\cdot y_{\text{N2}}}{M_{\text{air}}} \tag{7}$$

3.2. Cell voltage analysis

The cell voltage analysis is conducted using a comprehensive PEM fuel cell system simulation model implemented in Matlab/Simulink®. This in-house developed model is divided into five subsystems: the cooling circuit model, the anode model, the cathode model, the stack model, and the electrical model. For cell voltage analysis, only the cathode model and the stack model are considered.

The cathode model employs a semi-stationary 0D approach with the same simplifications as explained in section 3.1.

The stack model with the calculations assessing the impact of overvoltages with CEGR on cell voltage is performed using output parameters from the cathode model. These computations are grounded in a quasi-stationary, semi-empirical 0D approach, following the methodology outlined by Wallnöver-Orgis et al. [27].

According to this approach, the effective cell voltage U_{cell} is determined by the reduction of the open circuit voltage OCV due to the connection of an electrical load and the resulting irreversible losses.

These irreversible losses, also known as overvoltages, can be divided into activation overvoltage η_{act}, ohmic resistance overvoltage induced by electron resistance $\eta_{\text{ohm,el}}$, ohmic resistance overvoltage induced by proton resistance $\eta_{\text{ohm,mem}}$ and concentration overvoltage η_{conc}. The calculation of the cell voltage is shown in equation (8).

$$U_{\text{cell}} = \text{OCV} - \eta_{\text{act}} - \eta_{\text{ohm,mem}} - \eta_{\text{ohm,el}} - \eta_{\text{conc}} \tag{8}$$

The OCV is determined by the difference between the standard voltage E_N and the internal current density η_{int} using equation (9). E_N, calculated in equation (10),

Considers the deviation from the standard state of the standard voltage E^0, which is calculated based on the free Gibbs energy $\Delta_R G_m^0(T,p^0)$ in equation (11).

The internal overvoltage η_{int} is determined by employing equation (12), which represents the losses due to the diffusion of hydrogen to the cathode and oxygen to the anode, alongside the consequent side reactions.

$$\text{OCV} = E_{\text{N}} - \eta_{\text{int}} \tag{9}$$

$$E_N = E^0 - \frac{\text{R}\cdot T}{z\cdot\text{F}}\cdot\ln\left(\frac{p_{\text{H}_2}}{p^0}\cdot\sqrt{\frac{p^0}{p_{\text{O}_2}}}\right) \tag{10}$$

$$E^0 = -\Delta_R G_m^0(T,p^0)\,/\,(z\cdot\text{F}) \tag{11}$$

$$\eta_{\text{int}} = \frac{\text{R}\cdot T}{\alpha_{\text{O}_2,\text{red}}\cdot\text{F}}\cdot\ln\left(\text{i}_{\text{int}}\,/\,\text{i}_{0,\text{cat}}\right) \tag{12}$$

The activation overvoltage η_{act} arises from the restricted reaction rate on the platinum catalyst at both the anode and cathode. η_{act} can be computed using equation (13).

The ohmic overvoltage induced by electron resistance $\eta_{\text{ohm},el}$ can be determined using equation (14), while the ohmic overvoltage stemming from proton resistance $\eta_{\text{ohm,mem}}$ can be computed using equation (15).

$$\eta_{act} = \frac{\text{R}\cdot T}{\alpha_{\text{H}_2(\text{ox,red})}\cdot F}\cdot\text{arcsinh}\left(\frac{i}{2\cdot i_{0,\text{an}}}\right) + \frac{\text{R}\cdot T}{\alpha_{\text{O}_2(\text{ox,red})}\cdot\text{F}}\cdot\text{arcsinh}\left(\frac{i}{2\cdot i_{0,\text{cat}}}\right) \tag{13}$$

$$\eta_{\text{ohm,el}} = i\cdot\left(R_{BPP} + \frac{\delta_{\text{cat}}}{\sigma_{\text{elec}}} + \frac{\delta_{\text{an}}}{\sigma_{\text{elec}}} + \frac{2\cdot\delta_{\text{GDL}}}{\sigma_{\text{GDL}}} + 2\cdot R_{\text{c,GDL/BPP}}\right) \tag{14}$$

$$\eta_{\text{ohm,mem}} = (i\cdot\delta_{\text{mem}})\,/\,\sigma_{\text{mem}} \tag{15}$$

The concentration overpotentials η_{conc}, also referred to as transport losses, arise from non-uniform conditions within the porous structure of the electrode, leading to insufficient reactant concentrations on the catalyst. The concentration overpotential can be computed using equation (16). The empirical correction factor ξ_1 considers the variation in the exchange current density i_0 with decreasing oxygen concentration, the uneven current distribution within the porous electrode, and the low internal current density at high external current density [27].

$$\eta_{\text{conc}} = \xi_1\cdot\frac{\text{R}\cdot T}{4\cdot\text{F}}\cdot\ln\left(\frac{i_{\text{lim}}}{i_{\text{lim}} - i}\right) \tag{16}$$

4. Results and discussion

4.1. System architecture

The fundamental structure of an air supply system comprises an air filtration system, a (turbo-) compressor, a heat exchanger, a humidifier, a bypass valve for the humidifier, a bypass valve for the stack, cut-off valves for the stack, a pressure control valve, and a silencer.

As outlined by Kim et al. [18] there are two distinct approaches to implementing exhaust gas recirculation, depending on the utilization of an additional recirculation compressor and the point at which recirculated exhaust air is introduced into the fresh air stream.

The configuration involving a control valve and the introduction of recirculated air upstream of the air compressor presents several advantages. Notably, it eliminates the need to pressurize exhaust air to a higher level, thus obviating the necessity for an extra compressor. This translates to cost savings, reduced spatial requirements, and diminished mass. Moreover, precise regulation of exhaust gas volume is achievable through the control valve, enabling variable utilization of CEGR across all operational states.

However, one disadvantage of this configuration is the need for more complex water management. In the control valve configuration, cold ambient air encounters warm, humid exhaust air, leading to condensation that can potentially impair the functionality and longevity of the compressor. This can occur due to increased wear on the blades from droplet fogging and corrosion of the roller bearings. Therefore, it is crucial to remove liquid water from the system.

Conversely, managing humidity is less challenging in the architecture where recirculation occurs after the air supply compressor. Here, a larger quantity of vapour can be directly fed into the stack, making this architecture well-suited for self-humidification purposes.

However, as self-humidification is not the focus of this study and factors such as installation space, costs, mass, and controllability are of primary importance, the design without an additional recirculation compressor is favoured.

The defined system architecture includes the necessary CEGR components: a recirculation valve, a water separator, a collecting tank, and a diaphragm pump, as illustrated in Fig. 3.

At the core of the CEGR system lies the recirculation valve, pivotal in governing the proportion of fresh air to recirculated exhaust gas. Engineered as a 3/2-way valve, it facilitates a direct link between the two streams. Crucially, the valve must withstand deionized water with a low pH value (3–5). Further important specifications for the valve are detailed in Table 2.

Efficient water separation is imperative for the required water separator. It should achieve a separation efficiency of at least 90 %,

Table 2
CEGR valve specifications.

Variable	Value	Unit
Max. Flow rate	130	g/s
Media temperature	−30 to 85	°C
Relative humidity	5–100	%
Inlet pressure	657 to 3000	mbar(a)
Service life	>8000	h
Response time	<200	ms
Power supply	9–16	V
Air leakage	<1	g/s

effectively removing water droplets from the air mixture along with any water wall film or accumulation, all while maintaining minimal pressure loss. Centrifugal separators prove highly effective for the required mass flow.

However, removing the separated water presents challenges, as pressure losses from the air filter, pipework, and the CEGR valve result in negative pressure within the water separator relative to ambient pressure.

Consequently, a collection tank equipped with a level sensor and a pump for water discharge becomes necessary. Considering the low mass flow, high efficiency, and simple design requirements, a diaphragm pump is the preferred choice.

4.2. Gas composition

This section elucidates the computational findings regarding the gas composition of the stack supply air with exhaust gas recirculation. Fig. 4 illustrates the impact of the recirculation ratio X_{EGR} on the oxygen stoichiometry across a range from 0 % to 45 %.

In the upper graph, representing standard operation without recirculation $X_{EGR} = 0$ %, the oxygen stoichiometry aligns with the manufacturer's specified values. Upon activating exhaust gas recirculation, the oxygen stoichiometry diminishes, evidenced by the concurrent downward shift of the function graphs with increasing recirculation ratio. Notably, at a current density exceeding 0.5 A/cm^2, an oxygen stoichiometry of $\lambda_{O2} = 1$ is attained at a recirculation ratio of 45 %.

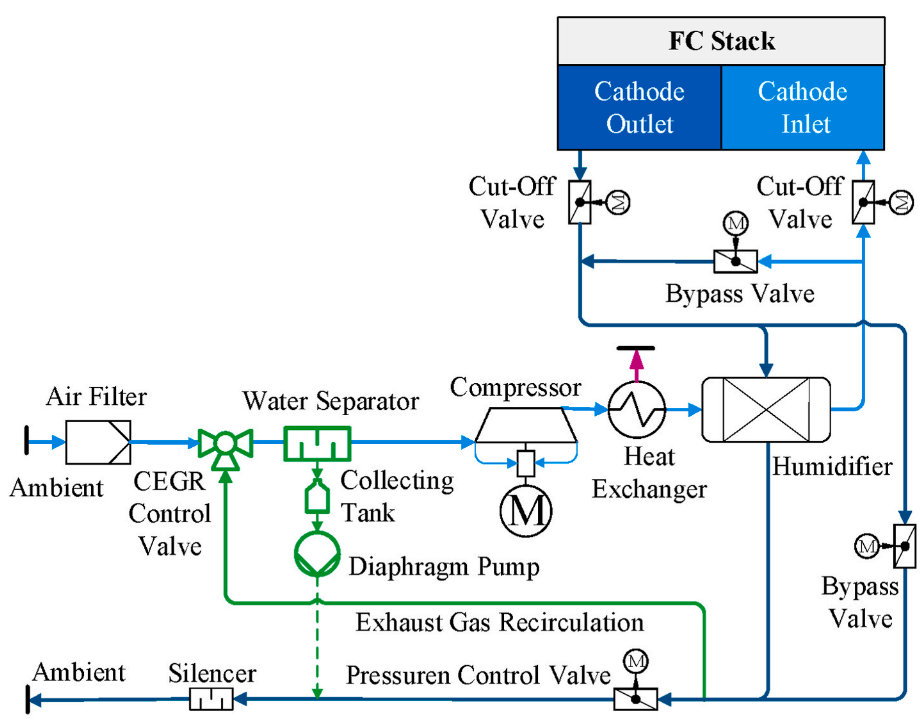

Fig. 3. Architecture of the air supply system with CEGR.

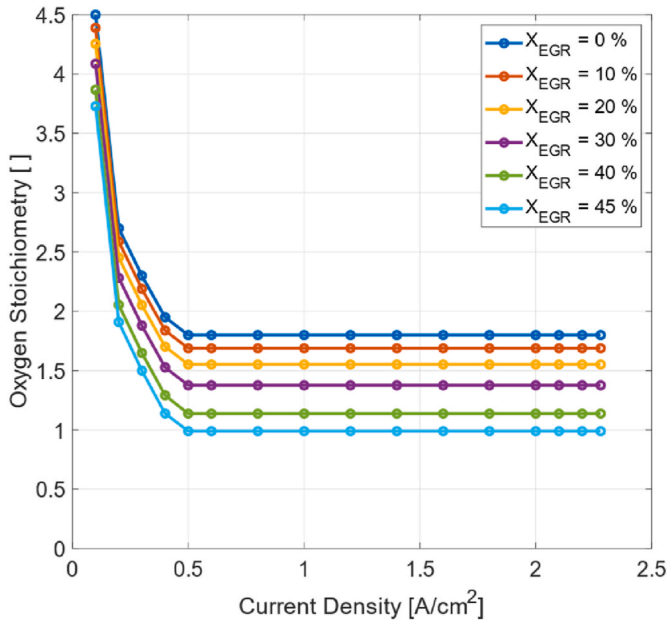

Fig. 4. Influence of the recirculation ratio X_{EGR} from 0 % to 45 % on the oxygen stoichiometry λ_{O2}.

Fig. 5 illustrates the segment corresponding to a current density of less than 0.4 A/cm^2. At lower current densities, a considerably higher degree of recirculation is necessary to reach an oxygen stoichiometry of $\lambda_{O2} = 1$. This heightened requirement can be attributed to the specified high supply air mass flow rate. For instance, at a current density of 0.1 A/cm^2, up to 82 % of the stack supply air must be sourced from recirculation to achieve $\lambda_{O2} = 1$. The correlation between oxygen stoichiometry and recirculation ratio is then examined individually for each current density.

Fig. 6 illustrates this relationship and serves as the foundation for subsequent control development, wherein the necessary recirculation ratio is derived from the required oxygen stoichiometry.

It is evident that at a low current density of <0.1 A/cm^2, the impact

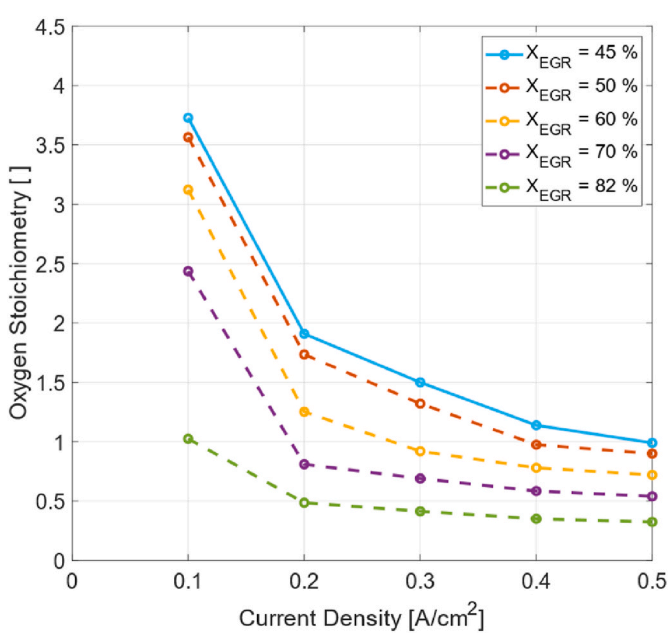

Fig. 5. Influence of the recirculation ratio X_{EGR} from 45 % to 82 % on the oxygen stoichiometry λ_{O2}.

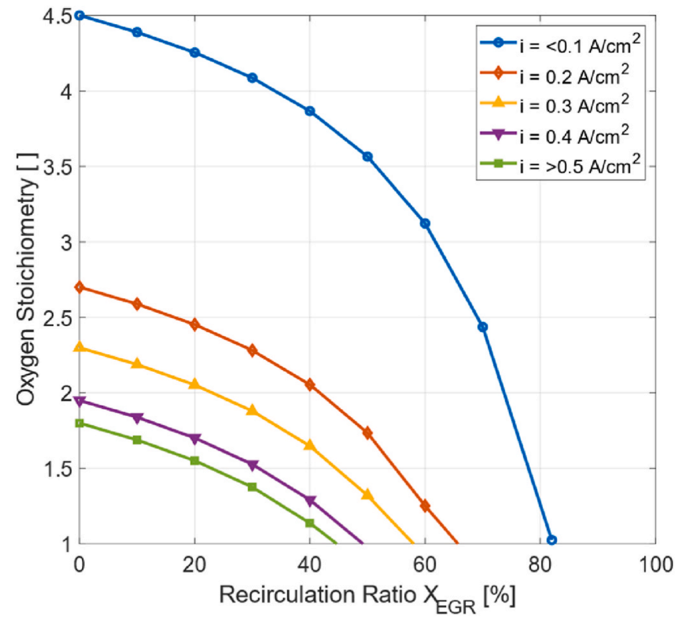

Fig. 6. Correlation between recirculation ratio X_{EGR} and oxygen stoichiometry λ_{O2} for individual current densities.

of a low recirculation ratio on oxygen stoichiometry is minimal. However, increasing the recirculation ratio at this current level results in a significant decrease in oxygen stoichiometry. Conversely, at current densities >0.5 A/cm^2, a less pronounced correlation between recirculation ratio and oxygen stoichiometry is observed.

4.3. Cell voltage

Fig. 7 a) illustrates the breakdown of the cell voltage without exhaust gas recirculation, considering an ambient temperature of 25 °C with a stationary observation of the cathode model. The cell voltage is broken down starting with the Nernst voltage, then subtracting internal, activation, electrical, membrane resistance, and concentration overvoltages. During normal operation, the cell voltage remains above 0.85 V at low current densities and decreases with increasing current density down to 0.59 V at 2.24 A/cm^2.

When incorporating exhaust gas recirculation at a controlled oxygen stoichiometry of $\lambda_{O2} = 1.1$, a notable decrease in cell voltage is noticeable. Fig. 7 b) demonstrates the breakdown of the cell voltage with CEGR and an oxygen stoichiometry of $\lambda_{O2} = 1.1$. It is noteworthy that the cell voltage can consistently be maintained below the critical limit of 0.8 V, and as the current density increases, the cell voltage diminishes, approaching zero.

Fig. 7 c) illustrates the cumulative difference between individual losses with and without exhaust gas recirculation. At low currents, this reduction in voltage is attributed to the decrease in the Nernst voltage and the concurrent increase in internal, activation, and concentration overvoltages. However, at higher currents, the rapid increase in concentration overvoltage primarily leads to the decline in cell voltage.

The correlation between cell voltage and X_{EGR} is mainly attributed to the decreasing oxygen partial pressure p_{O2}. Changes in stack supply air temperature are not considered in stationary operation. However, an increase in relative humidity contributes to an improvement in the membrane resistance, which results in a reduction in this overvoltage.

The rapid increase of concentration overvoltage η_{conc} is a result of the restricted concentration of reactants on the catalyst, triggered by the reduced oxygen partial pressure of the cathode supply air. η_{conc} steadily rises with X_{EGR} and i until it ultimately reaches the limiting current density i_{lim}, after which it diminishes towards zero.

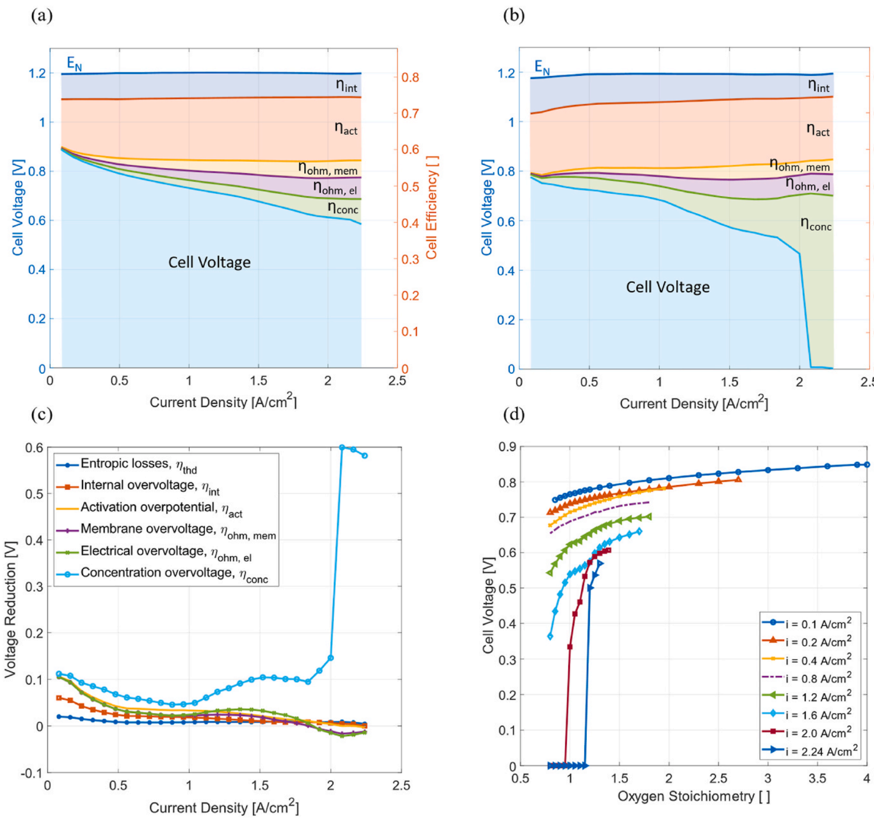

Fig. 7. Simulation results of the influence of CEGR on cell voltage. Breakdown of the cell voltage without CEGR is illustrated in (a), the breakdown of the cell voltage with CEGR at $\lambda_{O2} = 1.1$ in (b), the accumulated difference of the individual overvoltages from (a) and (b) is shown in (c) and the correlation between oxygen stoichiometry λ_{O2} and cell voltage for individual current densities in (d).

The correlation between oxygen stoichiometry and cell voltage can be derived for each current density. Fig. 7 d) illustrates this relationship for current densities ranging from 0.1 A/cm^2 to 2.24 A/cm^2. Particularly at high current densities, there is a significant decline in cell voltage with decreasing oxygen stoichiometry. At low current densities, there is a continual decrease in cell voltage as oxygen stoichiometry decreases. These correlations between cell voltage and oxygen stoichiometry hold significant importance for the ongoing analysis of exhaust gas recirculation in the context of control and operating strategy development. Based on these correlations, an effective control system can be devised to optimize idling and freeze starts.

4.4. Freeze start

A freeze start procedure incorporating CEGR aims to optimize the generation of waste heat. This optimization serves to accelerate freeze starts while concurrently mitigating the potential for heightened degradation or voltage instabilities.

Fig. 8 illustrates the potential waste heat generation of the fuel cell stack, once with an oxygen stoichiometry of $\lambda_{O2} = 1.1$ and once without exhaust gas recirculation. The figure shows that at $\lambda_{O2} = 1.1$ especially at high current densities, the potential for increased waste heat generation through elevation of the concentration overvoltage is evident. At a current density of 2 A/cm^2, the generated waste heat could be increased from

156 kW–180 kW (relative increase of +15 %) and at a current density of 2.08 A/cm^2, the generated waste heat could be increased from 164 kW to 275 kW (relative increase of +68 %).

However, operating the cell in the region of the limiting current density (greater than 2 A/cm^2) can cause irreversible damage to the fuel cell components. These.

At low system temperatures, the electrical input and output power of

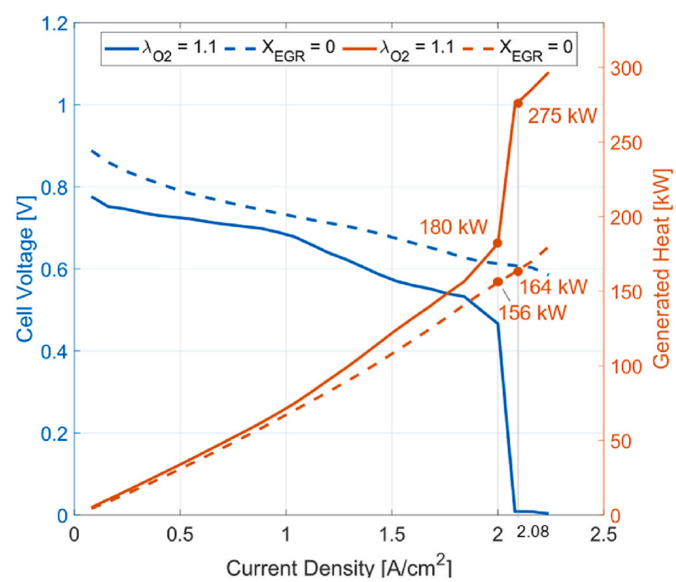

Fig. 8. Cell voltage and generated heat with ($\lambda_{O2} = 1.1$) and without ($X_{EGR} = 0$) CEGR conditions can lead to degradation of the catalyst and membrane [22].

the battery is limited, thus restricting the permitted FC electrical power output. The positive effect of CEGR is enhanced under this restriction. Due to the lower cell voltage, a higher current density can be maintained with the same power output as without CEGR, thereby reinforcing the generated waste heat. However, waste heat must be increased at a low current density, necessitating a lower oxygen stoichiometry. In this system, the power limit is set at 10 kW. This is representative for a freeze start in a vehicle where no driving power is requested.

A thermodynamic simulation with an active CEGR strategy investigates freeze starts from - 30 °C. In this simulation an oxygen stoichiometry below 1 is allowed to maintain the cell voltage at 0.2 V, ensuring the stack power does not exceed 10 kW, see Fig. 9. The model accounts oxygen stoichiometries below 1 by proportionally reducing the active cell area.

Results, shown in Fig. 10, indicate in the first 30 s no effect of exhaust gas recirculation at temperatures below 10 °C. This observation stems from the presumption that ice formation already reduces the active cell surface at subzero temperatures. However, once the ice melts, the influence of exhaust gas recirculation becomes significant.

The absence of the oxygen reactant on the cathode side results in 4.6 times faster heat-up to operating temperature compared to a system without exhaust gas recirculation. The required operating temperature of 60 °C can be reached in around 50 s, whereas without CEGR, a freeze start takes up to 230 s.

4.5. Idling

Two primary objectives are pursued during idling. Firstly, the aim is to reduce the fuel cell degradation, especially platinum oxidation, by operating at a cell voltage lower than 0.8 V and minimizing the number of start-stop cycles [22]. Secondly, hydrogen consumption is to be minimized thanks to the possibility of lowering the minimum stack power output.

Fig. 11 demonstrates the potential for power reduction during idle operation. The minimum current density is constrained by the critical voltage threshold of 0.85 V. Operating below this limit permits a decrease in the minimum idle current density, provided no other subsystem imposes limitations.

In the stationary simulation, a cell voltage below 0.8 V could consistently be attained up to a current density of 0.02 A/cm^2. This indicates, that the minimal stack power can be decreased from 9.2 kW to 1.7 kW, equaling the BoP consumption and therefore reaching an overall FCS net zero power output. This prevents the fuel cell from switching off when the battery is fully charged, thereby extending its useful lifetime.

In addition, hydrogen consumption is proportional to the current density. A reduction in current density from 0.12 A/cm^2 to 0.02 A/cm^2, representing an 83 % decrease, corresponds to a hydrogen saving of

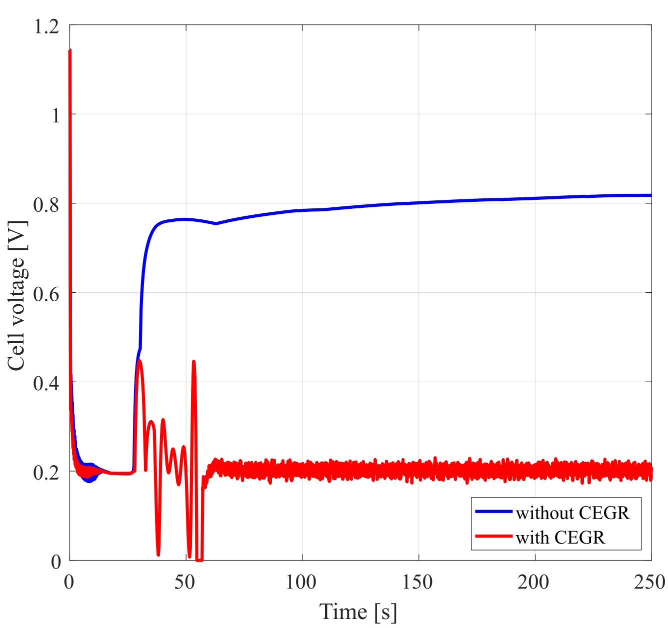

Fig. 9. Comparison of the cell voltage during a freeze start with and without exhaust gas recirculation.

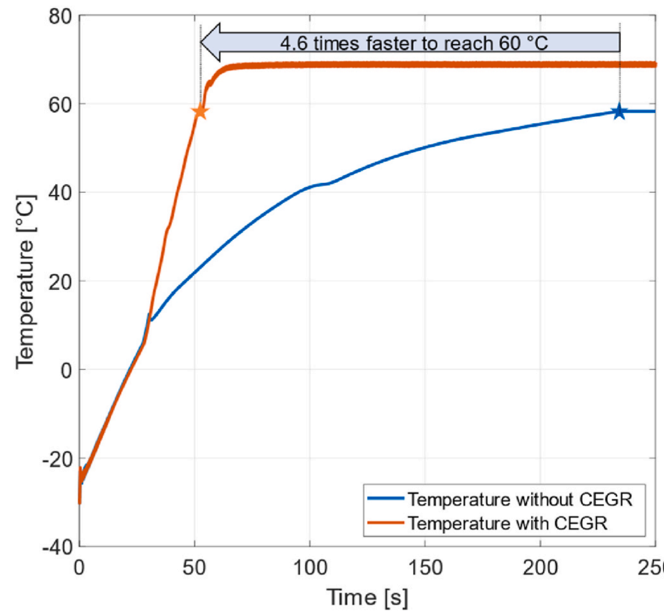

Fig. 10. Comparison of the freeze start-up time to operating temperature with and without exhaust gas recirculation.

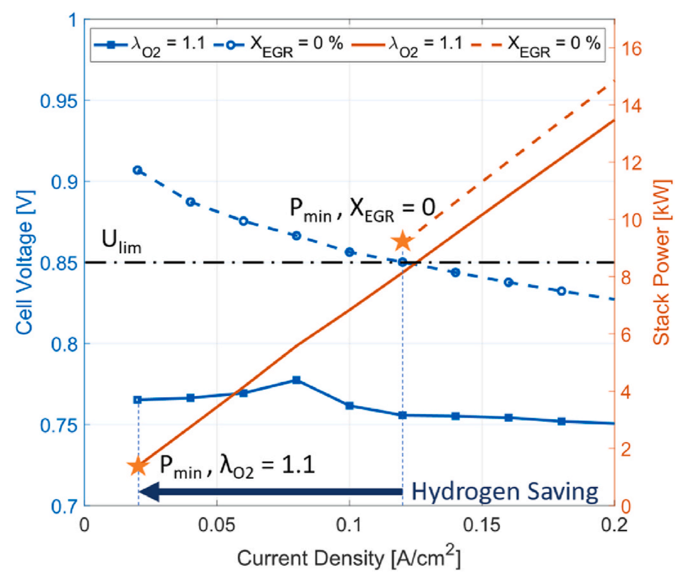

Fig. 11. Cell voltage and stack power with CEGR ($\lambda_{O2} = 1.1$) and without CEGR ($X_{EGR} = 0$) and additional labelling of the minimum permissible stack power for both cases and the hydrogen saving potential by reducing the current density.

0.094 g_{H2}/s.

4.6. Control system concept

The control concept of CEGR is similar to the approach outlined in Ref. [26] and is shown in Fig. 12. Depending on the strategy for idling or freeze starts, the fuel cell system's necessary current and voltage are specified and X_{EGR} is derived accordingly.

For a freeze start, the maximum output power, P_{max}, must be defined based on the system temperature and the vehicle's power requirements. In this study, P_{max} is set at 10 kW, assuming the vehicle is stationary during the freeze start, see equation (17). If the vehicle is in motion, the power requirement would be higher, thereby raising this power limit

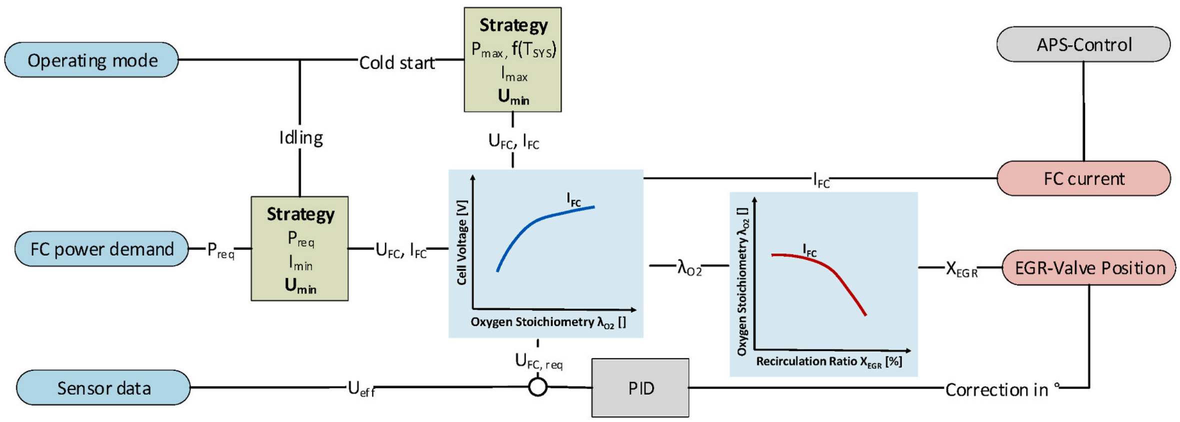

Fig. 12. Control of the exhaust gas recirculation system, consisting of an operating strategy, the simulation results from sections 4.2 and 4.3 and a PID controller for the cell voltage.

and amplifying the impact of CEGR. P_{max} and the temperature-dependent maximum current I_{max}, directly determine the cell voltage U_{min} to be achieved through CEGR, given in equations (18) and (19).

$$P_{max} = f\left(T, P_{req}\right) = 10\ \text{kW} \tag{17}$$

$$I_{max} = f(T) \tag{18}$$

$$U_{min} = \frac{P_{max}}{I_{max} \cdot A \cdot N} \tag{19}$$

During idling, X_{EGR} varies based on the power demand from the fuel cell and auxiliary units (BoP). It makes sense to reduce the electrical power by lowering the cell voltage when the battery is fully charged and the required power is up to 9.2 kW, which corresponds to the minimal idling power P_{min} without CEGR, shown in Fig. 11. If the vehicle power demand exceeds 9.2 kW, the system should operate at optimal cell efficiency without CEGR. Idle mode maintains a constant cell voltage U_{idle} of 0.79 V. Simulation results indicate that this voltage can be sustained with CEGR at any current up to 0.02 A/cm^2. This knowledge is used to determine the minimum current I_{min} in idling mode. The calculation of the required power is given in equations (20)–(22).

$$P_{req} = P_{vehicle} + P_{BoP} + P_{Bat} \tag{20}$$

$$U_{idle} = 0.79\ \text{V} \tag{21}$$

$$I_{min} = \frac{P_{req}}{0.79 \cdot A \cdot N} \tag{22}$$

The necessary oxygen stoichiometry λ_{O2} and recirculation ratio X_{EGR} are established by leveraging the defined currents and cell voltages outlined in the operating strategy. This process utilizes the simulation results obtained in both section 4.2 and section 4.3.

Using X_{EGR} in conjunction with the valve characteristics and the thermodynamic state variables of the fluids, the opening angle of the recirculation valve can be calculated. The recirculation valve is then controlled using a proportional-integral-derivative (PID) controller so that the desired cell voltage is achieved.

This control system requires further optimization to consider the influence of the back-pressure valve and the possible resulting resonance, as well as the dynamic effects of the cathode path and the related time delays.

4.7. Risk analysis

This section describes the potential risks associated with the introduction of CEGR. Table 3 provides an overview of these risks, accompanied by a brief description of their potential consequences and proposed solutions. Risks were identified based on typical operating conditions of PEM fuel cells found in the literature, supplemented by team brainstorming. A comprehensive analysis of risk assessment and risk management should be carried out as part of additional research efforts on this topic.

5. Conclusion

The findings of this study corroborate the functionality and effectiveness of the cathode exhaust gas recirculation (CEGR) system. The developed system has the potential to expand the operating range of the PEM fuel cell, reduce the consumption of hydrogen, and prolong the fuel cell's useful lifetime.

The defined system architecture focuses on controllability and includes a water management system. The simulation results highlight the detailed impact of exhaust gas recirculation on overvoltages and the consequent cell voltage.

In terms of freeze start capabilities, the CEGR system demonstrates the potential for higher waste heat generation, achieving a freeze start from −30 °C to operating temperature 4.6 times faster than a system without CEGR, in around 50 s. The system is particularly effective in reaching operating temperatures from above 0 °C.

In comparison to previous research on idle states, this study introduces innovative concepts for hydrogen savings and the generation of zero-idle power, thus avoiding the need to shut down the fuel cell. A reduction in hydrogen consumption of up to 83 % is achievable, with the minimum idling power reduced from 9.2 kW to 1.7 kW. This approach enhances efficiency and prolongs the fuel cell's lifespan.

The conducted risk analysis suggests areas for future research, emphasizing the need for closer examination of the limits and duration of oxygen starvation. Additionally, practical tests across the full current density range are required to validate the simulation results and gather detailed estimates of the service life extension.

In further project work, short stack tests are therefore planned to validate the results and record the simulated characteristic curves. Furthermore, comprehensive tests involving the entire fuel cell system will be undertaken to optimize control and operating strategies.

In conclusion, this study establishes a foundation for an optimized CEGR system designed for freeze start and idle conditions, opening new research fields, and providing impetus for further in-depth studies.

CRediT authorship contribution statement

Martin Aggarwal: Writing – review & editing, Visualization, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Daniel Valentin:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology,

Table 3
Risk analysis for CEGR.

Risk	Description	Consequence	Countermeasures
Local oxygen starvation	Local oxygen starvation due to low oxygen stoichiometry [35]	Degradation of the fuel cell	Electrochemical flow simulation, tests on a single cell and a short stack, state of health monitoring
Recirculating humid air	Liquid water and air with high relative humidity are recirculated	Shortening the service life of the compressor rotor and air-bearing	Water separators, flow simulations, coating and drying procedure during shut-down
Recirculation of acidic liquid	The recirculated water has an acidic pH value between 3 and 7 [36]	Increased corrosion of components in the fresh air path	Selection of material to resist acid conditions, nanoceramic electro-chemical oxidation (ECO)
Recirculation of hydrogen	Hydrogen is introduced into the exhaust gas through hydrogen-pumping and diffusion processes, after which it undergoes recirculation [37]	Degradation of the fuel cell, efficiency losses	Consideration in the operating strategy, cyclical flushing with fresh air
Heat dissipation in idle mode	CEGR generates additional waste heat	System overheating	Consideration in the design and control of thermal management
Interactions of the EGR valve in the cathode path	Interaction between the recirculation valve, the pressure control valve and the compressor speed	Vibrations, pressure surges, voltage peaks and oxygen shortages that lead to degradation or mechanical damage	Consideration in the control of the AP-system, practical tests at subsystem level
Lowering the cell voltage	Reduction of the cell voltage to a minimum	Not given functionality of the electrical components (DCDC) and voltage overlaps with the battery	Design of the power electronics for low voltages and overlapping voltage levels
Water discharge failure	Failure and leakage of the diaphragm pump, overfilling of the collection tank	Flooding and contamination of the cathode section	Monitoring the functionality of the diaphragm pump and level sensor in the water collection tank.

Investigation, Formal analysis, Data curation. **Samuele Cappelli:** Writing – review & editing, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Thomas Loidl:** Validation, Software, Investigation. **Patrick Pertl:** Writing – review & editing, Supervision. **Alexander Trattner:** Writing – review & editing, Supervision.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the corresponding author used OpenAI's ChatGPT-4 in order to improve language and readability. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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